

Question:

At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<think>

- At birth , the human brain shows a classic pattern : the thalamus , hypothalamus , and basal ganglia are among the most mature , whereas the frontal cortex , parietal cortex , visual system , cerebellum , and limbic structures show extensive myelination and white matter , increasing later in infancy and early childhood .

- The thalamus , often termed the “relay station” integrates and distributes sensory inputs to cortical areas and maintains motor integration . It matures much earlier than the cortex .

- This timing is reflected in neonatal electroencephalography and functional imaging : thalamic responses are mature and begin developing by birth ; cortical activation typically emerges later and shows smaller amplitude and slower maturation .

- Why the other options are less “least developed” at birth :

- Frontal lobe : begins myelination and significant maturation postnatally .

- Parietal lobe and visual system : exhibit strong myelination and maturation later , with prominent brain - visual reflexes at birth (Romberg , optokinetic nystagmus) .

- Cerebellum : begins developing but remains incomplete at birth ; appears reactive rather than attentive to visual stimuli .

- Limbic system and basal ganglia : also show relatively early maturation , but the thalamus is the most immature subsystem at birth .

- Clinical relevance : neonatal seizures are frequently associated with thalamic dysfunction due to incomplete myelination at birth , underscoring the thalamus’s relative immaturity at the start of life .

Therefore , the least developed part of the brain at birth is the thalamus .

Correct answer : B . thalamus

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<answer>

B . thalamus

</answer>